LAB 8 – Routing, NAT and PAT Configuration

Course: The Bits and Bytes of Computer Networking
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Date: April 20, 2026

1. Introduction

This lab aims to configure and validate communication between a private network and an external network using **static routing** and **Network Address Translation (NAT with PAT)**.

A real-world scenario is simulated in which a host from a private network accesses an external server, similar to Internet access.

2. Network Topology

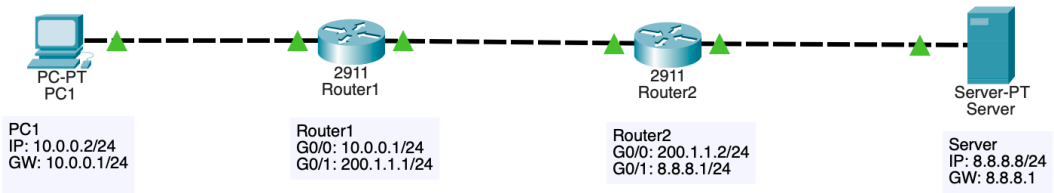
The network used in this lab consists of:

- One PC (client)
- Two routers (R1 and R2)
- One server

The connection is established as follows:

PC → R1 → R2 → Server

Figure 1 – Network Topology



Addressing Scheme

Device	Interface	IP Address
PC	—	10.0.0.2
R1	G0/0	10.0.0.1
R1	G0/1	200.1.1.1
R2	G0/0	200.1.1.2
R2	G0/1	8.8.8.1
Server	—	8.8.8.8

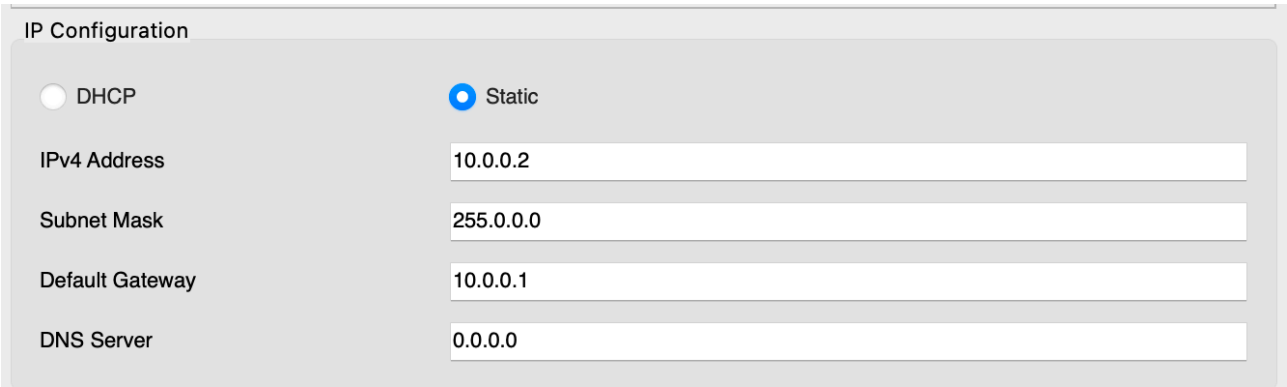
3. Network Configuration

3.1 PC Configuration

The PC was configured with the following parameters:

- IP Address: 10.0.0.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.0.0.1

Figure 2 – PC IP Configuration

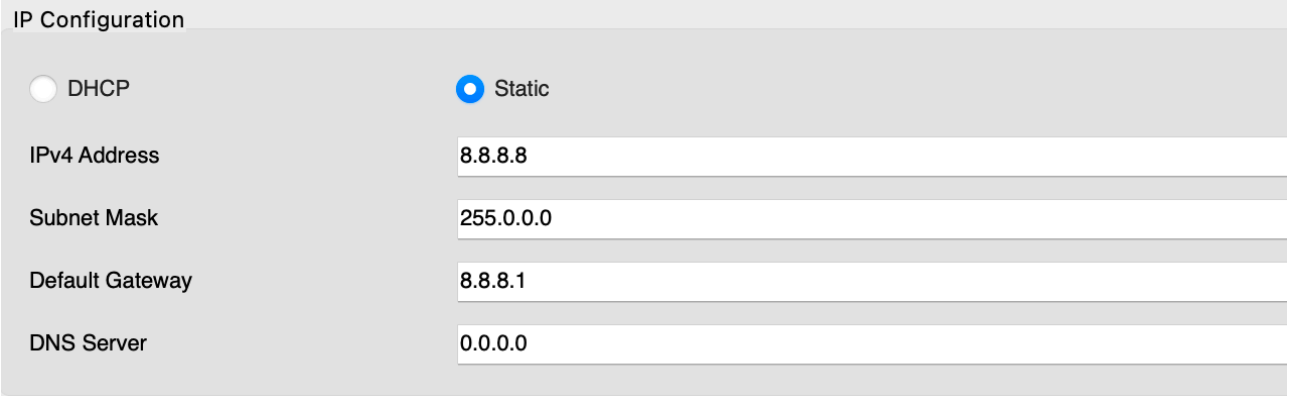


3.2 Server Configuration

The server was configured with:

- IP Address: 8.8.8.8
- Subnet Mask: 255.255.255.0
- Default Gateway: 8.8.8.1

Figure 3 – Server IP Configuration



Router Configuration

R1

```
interface g0/0
 ip address 10.0.0.1 255.255.255.0
 no shutdown

interface g0/1
 ip address 200.1.1.1 255.255.255.0
 no shutdown
```

R2

```
interface g0/0
 ip address 200.1.1.2 255.255.255.0
 no shutdown

interface g0/1
 ip address 8.8.8.1 255.255.255.0
 no shutdown
```

Figure 4 – R1 Interface Status

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     200.1.1.1       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Figure 5 – R2 Interface Status

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     200.1.1.2       YES manual up                    up
GigabitEthernet0/1     8.8.8.1         YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

4. Routing Configuration

To enable communication between networks, static routes were configured.

4.1 Default Route on R1

ip route 0.0.0.0 0.0.0.0 200.1.1.2

This route allows R1 to forward all unknown traffic to R2.

4.2 Return Route on R2

ip route 10.0.0.0 255.255.255.0 200.1.1.1

This configuration allows R2 to send return traffic back to the internal network

Figure 6 – Routing Table on R1

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.1.1.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
     200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/24 is directly connected, GigabitEthernet0/1
L       200.1.1.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 200.1.1.2

Router#
```

Figure 7 – Routing Table on R2

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     8.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       8.8.8.0/24 is directly connected, GigabitEthernet0/1
L       8.8.8.1/32 is directly connected, GigabitEthernet0/1
     10.0.0.0/24 is subnetted, 1 subnets
S       10.0.0.0/24 [1/0] via 200.1.1.1
     200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/24 is directly connected, GigabitEthernet0/0
L       200.1.1.2/32 is directly connected, GigabitEthernet0/0

Router#
```

5. NAT and PAT Configuration (R1)

To allow communication between the private network and the external network, **NAT with PAT** was configured.

5.1 Defining the Internal Network

access-list 1 permit 10.0.0.0 0.0.0.255

This ACL identifies the private IP addresses that will be translated.

5.2 NAT Interface Configuration

interface g0/0
 ip nat inside

interface g0/1
 ip nat outside

- **Inside → Private network**
- **Outside → External network**

5.3 Enabling PAT

ip nat inside source list 1 interface g0/1 overload

This allows multiple connections to share a single public IP address using port translation (PAT).

Figure 8 – NAT Configuration

```
Router#show running-config | include access-list
access-list 1 permit 10.0.0.0 0.0.0.255
Router#show running-config | include nat
 ip nat inside
 ip nat outside
ip nat inside source list 1 interface GigabitEthernet0/1 overload
Router#
```

6. Verification

6.1 Connectivity Test

From the PC:

ping 8.8.8.8

Figure 9 – Successful Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

6.2 NAT Verification

`show ip nat translations`

Figure 10 – NAT Translation Table

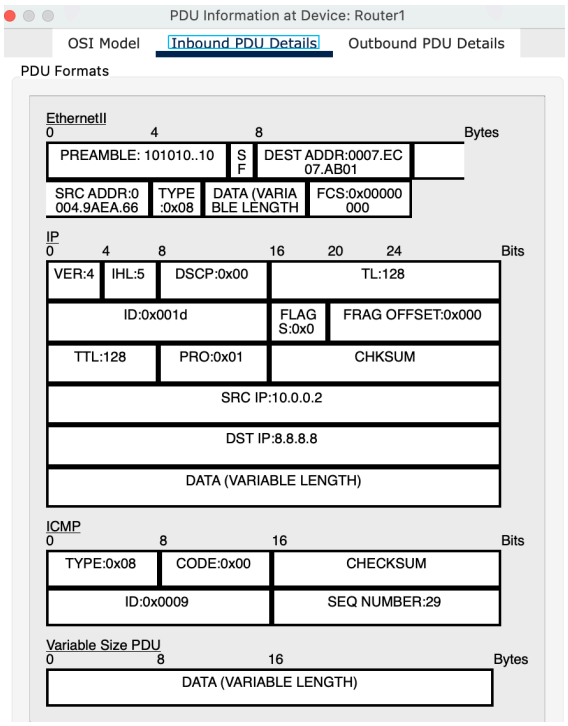
```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.1.1.1:21      10.0.0.2:21        8.8.8.9:21         8.8.8.9:21
icmp 200.1.1.1:22      10.0.0.2:22        8.8.8.9:22         8.8.8.9:22
icmp 200.1.1.1:23      10.0.0.2:23        8.8.8.9:23         8.8.8.9:23
icmp 200.1.1.1:24      10.0.0.2:24        8.8.8.9:24         8.8.8.9:24
icmp 200.1.1.1:25      10.0.0.2:25        8.8.8.8:25         8.8.8.8:25
icmp 200.1.1.1:26      10.0.0.2:26        8.8.8.8:26         8.8.8.8:26
icmp 200.1.1.1:27      10.0.0.2:27        8.8.8.8:27         8.8.8.8:27
icmp 200.1.1.1:28      10.0.0.2:28        8.8.8.8:28         8.8.8.8:28
```

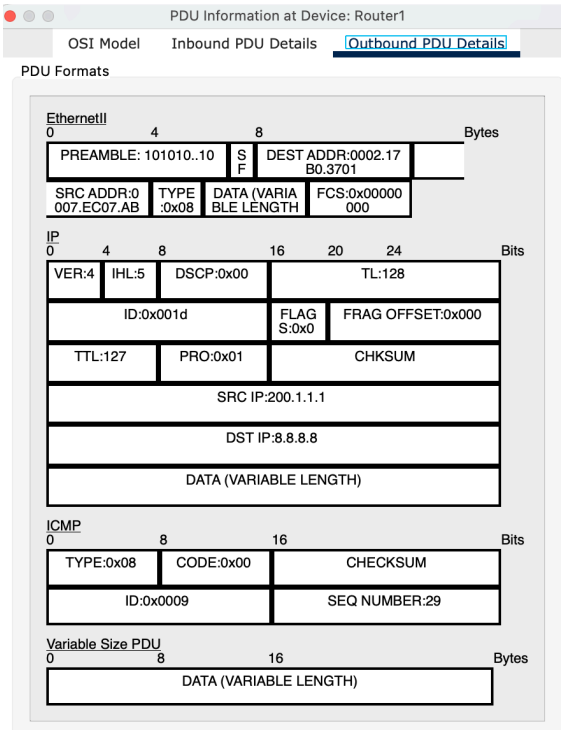
This table shows the translations performed by R1, where the private IP address (10.0.0.2) is translated into the public IP address (200.1.1.1).

It also demonstrates the use of **PAT (Port Address Translation)**, as multiple connections share the same public IP address while being differentiated by port numbers.

6.3 (Optional) Simulation

Figure 11 – Simulation Mode





In the **Outbound PDU Details**, the source IP address changes from **10.0.0.2 to 200.1.1.1**, demonstrating NAT operation.

7. Analysis

The communication flow is as follows:

- The PC sends traffic to the server
- R1 translates the private IP into a public IP using NAT
- R2 routes the traffic to the server
- The server responds successfully

The correct operation of the following was verified:

- Routing
- NAT
- PAT

8. Conclusion

This lab demonstrated how a private network can access an external network using routing and address translation.